

Introduction

Understanding customer behavior and preferences is important for any media company that needs to promote its content and services to its audience. In this project, we want to group customers for a digital magazine, using behavioral and article data to get insights into customer needs and preferences. We are using two data sets: *behavioral data*, containing information on customers' browsing habits, demographics, and engagement metrics, and *article data*, giving the topics of interest to each customer over the past three months. By using clustering, we can group similar customers together, allowing the company to better understand its customers base. This can help enhance customer satisfaction, engagement, and develop marketing strategies.

Methods

For both data sets I use Hierarchy Agglomerative Clustering. For the first data set, I started by separating continuous and categorical data. I z scored with each type being put into their respective encoder. I tested each model and evaluated their performances by using a silhouette score as it tells us how similar data points are to their own cluster compared to the other clusters. HAC performed the best out of each model with their respective best hyperparameters, with the closest value to one (0.61), so I decided to use it. Next, I made a PCA pipeline which reduces variables while preserving the most important information. After fitting the PCA on the scaled data, I created a scatterplot of the clusters by plotting the first Principal Component (PC1) on the x-axis and the second Principal Component (PC2) on the y-axis, colored by their respective clusters. This visualization allowed me to observe the separation of clusters in a lower-dimensional space. After clustering and visualizing the data, I summarize the characteristics of each cluster by calculating the mean values of the original features within each cluster, and putting them all in a table.

For the second data set, I used the same model but with the affinity set to cosine since it uses counts and the linkage set to average since the data is very spread out. However, initially I didn't set a number of clusters, so I could create a dendrogram to analyze the data for the number of clusters needed. After generating the dendrogram, I made the threshold for clusters at 0.45, since any more would have had data points too far apart in the same cluster. As a result, the number of clusters was 5. I refit the HAC model to fit for 5 clusters and generate the labels to put data points in clusters. Finally, like in number 1, I generated a table with the mean values of all of the features for each cluster.

Results

In the Behavioral data, my models performed very well as the visual of the PC1 vs PC2 scatterplot shows clear clusters of the data points. Based on the clustering analysis of the mean table, there are multiple conclusions that could be possible: Cluster 1 represents customers with higher incomes, longer browsing times, and higher engagement with ads. This can represent more wealthy people that have more free time. In contrast, Cluster 2 consists of

budget-conscious customers with lower incomes and shorter browsing sessions. Cluster 0 includes customers with relatively higher ages and incomes, along with moderate browsing times and engagement with ads. This could represent retired individuals with more money and time than others. Cluster 3 comprises customers with lower incomes, shorter browsing times, and minimal engagement with ads. These clusters can exhibit differences in behavior, indicating their coherence and effectiveness. Leveraging this insight, the company can tailor marketing strategies to target each segment more effectively. For example, Cluster 1 may respond well to premium offerings and personalized ads, while Cluster 2 and 3 may prefer cost-effective promotions. By catering to the needs of each group of customers, the company can improve customer satisfaction and optimize its marketing efforts.

Based on the data from the average tables from the Article data set, we can see various trends within each cluster based on their interests in the topics. Cluster 0 appears to have individuals with an interest in Technology and AI, as shown by the highest average values for these topics compared to other clusters. Cluster 1 shows a more balanced interest across multiple topics, including Celebrity, Fashion, and SelfHelp. Cluster 2 seems to be dominated by Science and Technology, while Cluster 3 exhibits a strong inclination towards Fashion and Celebrity-related content. Finally, Cluster 4 stands out with an exceptionally high interest in Productivity, suggesting a focus on personal development and efficiency. The clustering analysis reveals distinct customer segments based on behavior and interests. The models performed well, given by clear clusters in the scatterplot and significant differences among clusters. I would recommend the CEO to create tailored marketing strategies for each group of people. Future considerations can be assessing the long-term impact of targeted strategies on business performance.

Discussion/Reflection:

Conducting the clustering analyses provided valuable information for customer grouping for company usage. The models performed well, shown by clear clusters in the scatterplots and differences among values in the clusters. It was interesting to see the different customer groups based on behavior and interests, ranging from wealthier individuals with high engagement to budget-conscious users with shorter browsing sessions. Tailoring marketing strategies to each group's preferences and needs could enhance customer satisfaction and optimize marketing efforts. Moving forward, it would be beneficial to assess the long-term impact of these targeted strategies on business performance. Overall, this analysis underscores the importance of understanding customer behavior and preferences in developing effective marketing strategies for media companies.

